

Setting the terms by slice

We held the anti-subspace weight high on the embedding and block 1 only, and at the recipe level on the pulled blocks. This kept most of the selectivity of a high hold everywhere, and it gave back the removal that the high hold cost. We also pooled the pull over slices as well as positions. On most seeds the pull then settled on the last block. But on one seed it settled on a position the prediction does not use, and there the op could not be removed. On one of the block-4 seeds, a softer pool over slices let the earlier blocks align too, and removal and spill stayed about where they were.

Removal. How much of difference the edit takes out...

Spill. The largest net drop in score on another op under the edit....

[The late-pull experiment](#) pulled `difference` toward e_1 at blocks 2 to 4 only, with the anti-subspace term on every slice (the `deep` arm). Holding the anti weight at 0.2 for all of training, in place of the recipe hold of 0.03, kept the color embedding table and block 1 off e_1 on the other ops and brought the spill within the criterion on every seed. But it removed about a tenth less of the op.

Anti-subspace term. The companion to the pull...

A second review of those runs found that the hold also presses on the pull at the pulled blocks, which may account for the lost removal. So this report sets the two terms by slice, in two arms of 4 runs each at 400 epochs:

- `split-anti`: the `deep` pull, with the anti weight held at 0.2 on the embedding and block 1 and at the recipe hold of 0.03 on blocks 2 to 4.
- `pool-slices`: the same anti weights, with the pull on blocks 1 to 4 pooled over slices as well as positions, so that it may choose where along the stream to align.

A second round asked whether a softer pool over slices would settle less of the pull on block 4. It adds 4 runs of `pool-slices` at seed 701, with the pool over slices at a temperature of its own.

Observations

- [Removal and spill \(E1\)](#): with the anti weight split by slice, removal is back to that of `deep` at the recipe hold, and the spill stays within the criterion at every dose on 3 of 4 seeds. The fourth spills a little past it at the full dose only.
- [What the anti weight presses on \(E2\)](#): the split keeps the alignment at the fitting example answers where the recipe hold has it, and keeps block 1 and the color embedding table nearly as far off `e1` on other ops as the high hold does.
- [Where the pooled pull settles \(E3\)](#): pooled over slices, the pull settles on block 4 on three seeds, and there the edit removes the op about as selectively as the split does. On seed 700 the pull settled on the query answer alone, a position the prediction does not read, and the op is not removable there.
- [A softer pool over depth \(E4\)](#): on seed 701, softening the pool over slices leaves block 4 where it was and lets blocks 1 and 2 align about as far as in `split-anti`. Removal and spill move around from one temperature to the next with no trend, and every run stays within the criterion.

Removal. How much of difference the edit takes out...

Spill. The largest net drop in score on another op under the edit....

Alignment. How far a state points along `e1`...

Fitting example. An example whose answer is one that difference could give on its operands, whatever op the context is under.

Scope

This is an exploratory study, with no preregistration and no gate. Four seeds per arm resolve the large differences between the late-pull holds, but not whether a single run past the criterion is typical. E4 has one run per temperature, so it can show a change in where the pull settles, but not a difference in removal of the size that seeds differ by.

Removal. How much of difference the edit takes out...

Each run shares its seed, and so its initialization, batches and label draws, with a late-pull `deep` run at each hold and an ex-2.2.23 control. So a difference between arms at one seed comes from the change of terms rather than the initialization or the batches.

The anchor weight (0.1), τ (0.1), the label, the corpus and the schedules are the recipe values. Both arms keep the recipe anti schedule: 0.25 at the start on every slice, annealing to its hold by epoch 360, then sharing the anchor anneal at the end. The anti weight is now written as an absolute number rather than a ratio to the anchor weight, which gives the same schedule here.

The measurements

The measurements are those of the late-pull experiment, with the dose scoring of ex-2.2.22 added for every run and its twins:

- Removal: the net drop in the score on `difference` under the edit, as a share of the way from the clean score to the target null. The edit projects e_1 out of the state at every position, on a set of slices, and at a dose of a quarter, a half, three quarters, or all of the e_1 component.
- Spill: the largest net drop on another op. Both are net of the control at the same seed, and the selectivity criterion is a spill of at most 0.02 at every dose.
- The alignment α with e_1 at the example answers, the positions where the anchor sits in this grammar, along training at every slice.
- The task score of each op, as expected exact match on held-out contexts, against the control at the same seed.

Removal. How much of `difference` the edit takes out...

Target null. The score an ideal predictor would get on `difference` contexts if it had lost that op and answered from the remaining ones.

Spill. The largest net drop in score on another op under the edit....

Alignment. How far a state points along e_1 ...

Expected exact match. The task score, as the chance that a drawn answer agrees with a true one....

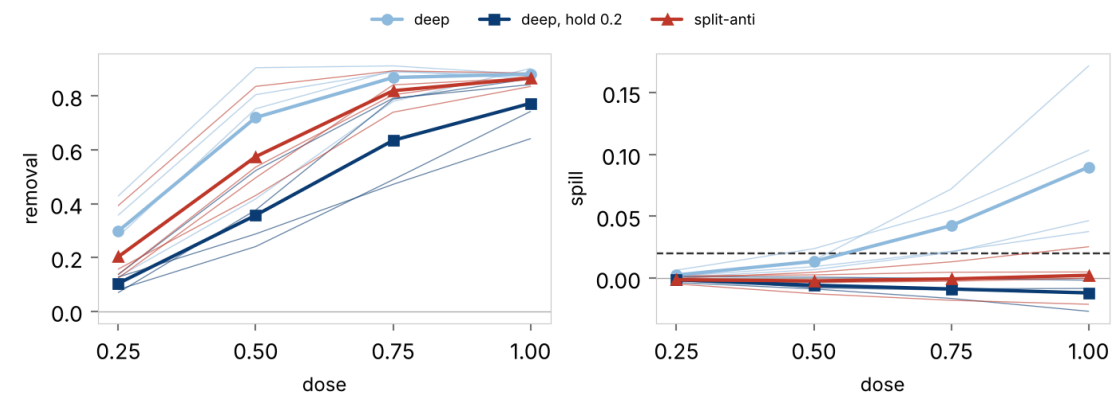
Removal and spill (E1)

If the high hold costs removal by pressing on the pull at blocks 2 to 4, holding it high only on the embedding and block 1 should give the removal back. If the stand-ins live at those first two slices, the spill should also stay low. The main measure is removal at the full dose, with the spill at every dose beside it.

Removal. How much of difference the edit takes out...

Stand-in. Something token-level that satisfies the pull where no contextual concept can...

Spill. The largest net drop in score on another op under the edit....



Removal and spill by dose, the edit at every position and slice. One hairline per run and the seed mean bold. The dashed rule is the selectivity criterion. `deep` and `deep, hold 0.2` are the late-pull runs at the same seeds.

Both hold. On every `split-anti` run, removal grows with the dose to about what `deep` removes at the recipe hold (the table below has the ranges). The spill stays within the criterion at every dose on 3 of 4 seeds. On seed 701 it passes the criterion at the full dose only (0.025, on `mix`), the same seed and op where `deep` at the recipe hold spilled least. The high hold everywhere is the only arm within the criterion on every seed.

arm	removal	within criterion	spill, any dose	raw spill	landing	task change
<code>deep</code>	0.88 (0.86 to 0.90)	0 of 4	0.090 (0.038 to 0.171)	0.122 (0.048 to 0.207)	0.47 (0.42 to 0.50)	-0.003 (-0.006 to 0.001)
<code>deep, hold 0.2</code>	0.77 (0.64 to 0.86)	4 of 4	-0.001 (-0.004 to 0.001)	0.006 (0.003 to 0.014)	0.48 (0.41 to 0.55)	-0.005 (-0.018 to 0.003)
<code>split-anti</code>	0.87 (0.84 to 0.88)	3 of 4	0.006 (-0.005 to 0.025)	0.015 (0.001 to 0.036)	0.49 (0.44 to 0.51)	-0.006 (-0.022 to 0.003)
<code>pool-slices</code>	0.63 (-0.04 to 0.90)	3 of 4	0.006 (-0.007 to 0.034)	0.029 (0.011 to 0.061)	0.36 (0.04 to 0.51)	-0.004 (-0.011 to 0.003)

Removal, spill, landing and task, seed mean and range over 4 runs. Removal is at the full dose; "spill, any dose" is the largest net spill over the four doses, and "within criterion" counts the runs where it is at most 0.02. The raw spill is the largest drop on another op at the full dose without subtracting the control. The landing is the share of the distance to the target null that the full edit covers, as total variation on `difference` contexts. The task change is the mean score over ops, less that of the control. `pool-slices` is the subject of E3.

The edit lands about as close to the target null in `split-anti` as in both `deep` arms. So the extra removal seems to be the op itself going away, rather than answer mass scattered onto other ops.

Target null. The score an ideal predictor would get on `difference` contexts if it had lost that op and answered from the remaining ones.

The task change is small, as in `deep`. One `split-anti` run (seed 702) has the lowest task score against its control of any run so far. The loss is spread across all ops rather than concentrated on `difference`, which suggests seed variation more than a cost of the terms. Four seeds cannot separate the two.

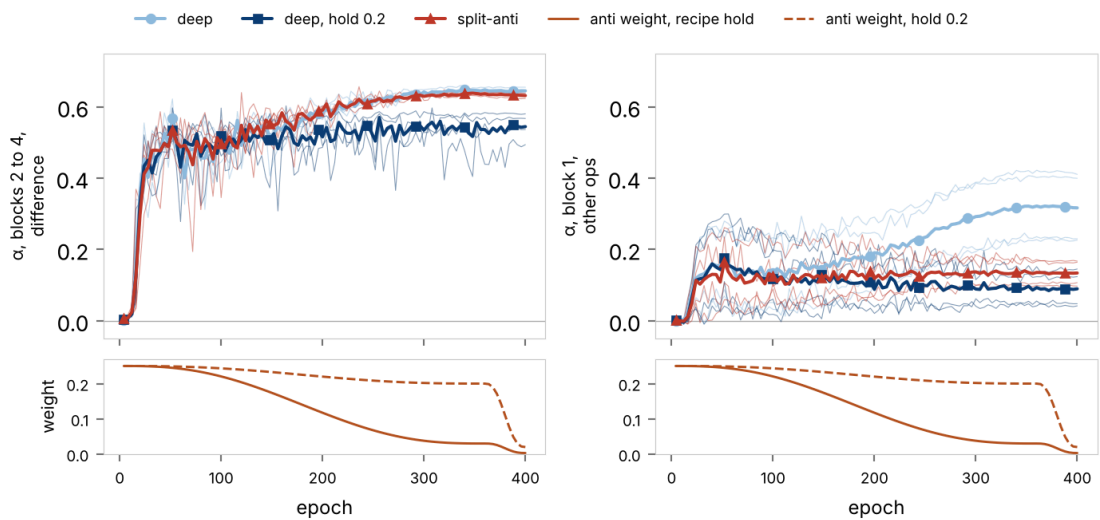
So on four seeds the split gets nearly all the selectivity of the high hold with none of its cost in removal. Which slices the edit has to cover is unchanged from late pull: as in `deep`, editing blocks 2 to 4 alone removes nearly everything on two seeds and less than half on the other two, and editing block 1 alone removes most of the op on every seed.

What the anti weight presses on (E2)

Did the removal come back for the reason we expected, the alignment at the pulled blocks restored with block 1 and the table still off e1 on the other ops? The main measure is α at the example answers: of difference contexts at blocks 2 to 4, and of the other ops at block 1.

Removal. How much of difference the edit takes out...

Alignment. How far a state points along e1...



Alignment at the example answers along training. Left: α at the example answers of difference contexts, averaged over blocks 2 to 4. Right: α at block 1 at the example answers of the other ops. One hairline per run and the seed mean bold. The strip below is the anti weight: deep at the recipe hold (solid) and at 0.2 (dashed). split-anti follows the dashed line on the embedding and block 1 and the solid line on blocks 2 to 4.

It did. At blocks 2 to 4, split-anti follows deep at the recipe hold: the alignment reaches about a half early in training and climbs through the second half, where the high hold everywhere holds it flat. At block 1 on the other ops it follows the high hold, and stays low where deep drifts up through the second half.

arm	fitting, difference	non-fitting, difference	fitting, other ops	block 1, other ops	slope in the table
deep	0.96 (0.95 to 0.96)	0.11 (0.09 to 0.14)	0.92 (0.90 to 0.94)	0.32 (0.23 to 0.41)	-0.26 (-0.30 to -0.18)
deep, hold 0.2	0.84 (0.76 to 0.90)	0.02 (0.01 to 0.04)	0.74 (0.66 to 0.83)	0.09 (0.04 to 0.15)	-0.04 (-0.12 to 0.01)
split-anti	0.94 (0.93 to 0.95)	0.09 (0.06 to 0.12)	0.90 (0.89 to 0.93)	0.13 (0.10 to 0.17)	-0.08 (-0.11 to -0.03)

Where the anchor sits at the end of training, seed mean and range over 4 runs. The first three columns are α at the example answers averaged over blocks 2 to 4, on held-out contexts, split by whether the example fits difference (whether that op could give the answer it shows) and by the op of the context. "Block 1, other ops" is α at block 1 at the example answers of the other ops, on the probe contexts. The slope is the lean of the color embedding table: the e1 component of the color embeddings against lightness. The controls are at about 0.04 in the first three columns and lean either way, up to about ±0.25.

Split by whether an example fits difference, the alignment at the fitting answers is back to nearly the recipe-hold level, so the pressure that the high hold put on the pulled blocks has been lifted.

The non-fitting answers are back near the recipe-hold level too. That alignment was never about the op: the high hold everywhere had cleared it and the split leaves it in place, but it does not show up in the spill.

Spill. The largest net drop in score on another op under the edit....

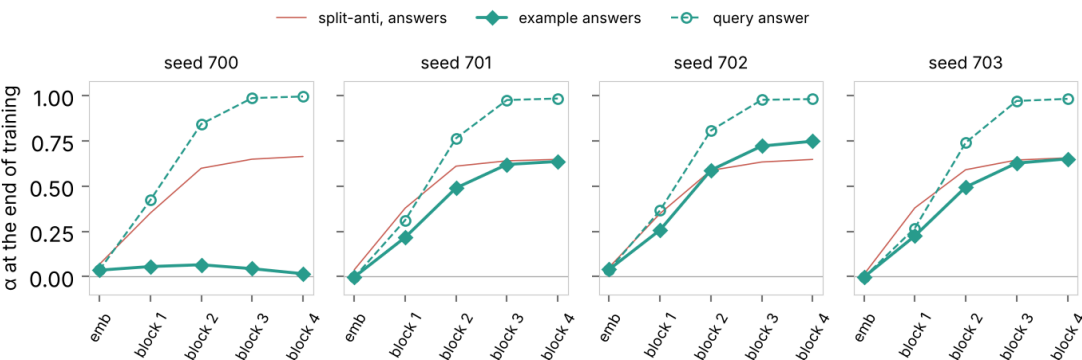
The color embedding table leans about a third as much as in deep, a little more than at the high hold, and on every seed it leans the way the anchored runs lean. Block 1 on the other ops is a little above the high hold too. Both fit the small spill that remains on one seed.

Where the pooled pull settles (E3)

Pooled over slices as well as positions, the pull only needs each labeled context to align at one place along the stream. Each place gets a share of the pull that grows steeply with how well aligned it already is, so the pull should settle on whichever slice aligned first.

Which slice is that, and does the edit at every slice still remove the op? The main measure is α at the example answers at each slice.

Alignment. How far a state points along e1...



Where the pooled pull settles, by seed. α at the end of training on *difference* probe contexts, at each slice: at the example answers (filled markers) and at the query answer (open markers), for *pool-slices*. The hairline is *split-anti* at the example answers, at the same seed.

On three seeds the alignment is largest at block 4, and the profile over slices looks much like that of *split-anti*: low at block 1 and rising to block 4, with blocks 2 and 3 nearly as high. Block 1 aligns less than in *split-anti*. On those seeds the spill is within the criterion at every dose, and the edit removes the op as well as in *split-anti* on two of them and about a tenth less on the third (seed 702).

Spill. The largest net drop in score on another op under the edit....

seed	largest α at answers	removal	block 1 only	blocks 2 to 4 only	spill, any dose
700	block 2	-0.04	0.04	0.02	0.034
701	block 4	0.90	0.58	0.94	-0.007
702	block 4	0.76	0.78	0.01	0.001
703	block 4	0.89	0.76	0.93	-0.004

The pooled pull, by seed. The slice with the largest α at the example answers of *difference* probe contexts, among blocks 1 to 4. Removal at the full dose of the edit at every slice, then of the edit on block 1 alone and on blocks 2 to 4 alone. The spill is the largest net spill over the four doses of the edit at every slice.

On seed 700 the example answers stayed near zero at every slice, and the edit removes nothing. What aligned instead is the query answer (the state at the answer token of the query, which has seen the answer and so can show the op), reaching α of 1 at blocks 3 and 4. That position aligns in every arm, since it is part of every labeled context, but in *pool-slices* on this seed it took the whole of the pull.

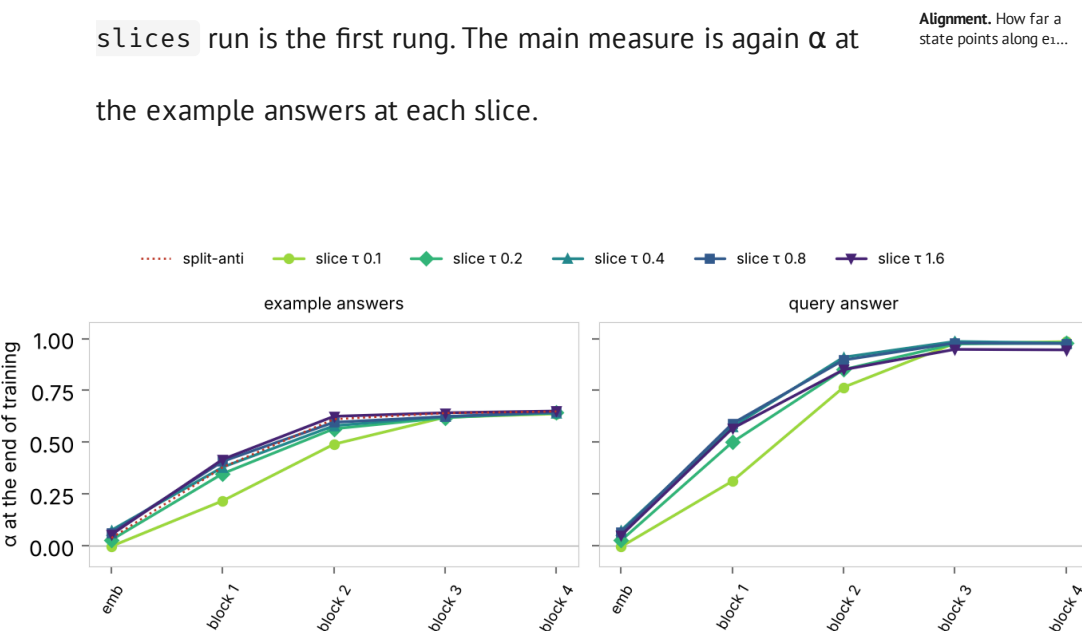
Along training the example answers rose to about a fifth early on, then fell back as the query answer took over. The prediction of the query answer is made at the query = , before that position, so an edit there cannot reach it.

This is the concentration the pooling was expected to bring, landing on a position no one meant it to use. The arm changes two things against *split-anti*, since block 1 joins the pulled slices as well as the pull being pooled; but block 1 took almost none of the pull on this seed, so the pooling is the likelier cause.

The per-slice pool includes the query answer too, and in every arm it aligns there about as far at blocks 2 to 4 as here. Why no per-slice run lost its example answers the same way, we can't say from these runs, and with one run in four it may be chance which position gets ahead.

A softer pool over depth (E4)

With the slices pooled at the recipe τ , the pull may put nearly all its weight on whichever slice is ahead, and on three seeds that was block 4. A softer pool over slices spreads the pull back over depth: at a high enough temperature it is the mean of one pool per slice. So we trained the `pool-slices` arm again at seed 701, a seed where it settled on block 4, with the pool over slices at 0.2 to 1.6, each double the last, while the pool over positions stays at the recipe τ . At the recipe τ the two pools are the one joint pool of E3, so the `pool-slices` run is the first rung. The main measure is again α at the example answers at each slice.



Where the pull settles as the pool over slices softens, seed 701. α at the end of training on `difference` probe contexts at each slice: at the example answers (left) and at the query answer (right), one line per temperature of the pool over slices, from the recipe τ (`pool-slices`, lightest) to the softest. The dotted line on the left is `split-anti` at the same seed.

Softening the pool did not take the alignment away from block 4: it stays at about 0.64 at every temperature, the level `split-anti` has it at too. What changes is the earlier blocks. At the recipe τ , block 1 is at 0.22 and block 2 below the blocks after it. From the first softer step on, both rise, and the profile over slices looks like that of `split-anti` at this seed. So on this seed the pool over slices at the recipe τ had not latched onto block 4 so much as held back the blocks before it, and a softer pool lets them align too.

The query answer shifts the same way, with blocks 1 and 2 higher at the softer temperatures. It aligns at blocks 2 to 4 at every temperature, as in the other arms, and the example answers keep their alignment, so this seed shows nothing of the seed 700 outcome. One seed cannot say whether a softer pool would have kept that seed on the example answers.

slice τ	α , block 1	α , block 4	removal	block 1 only	blocks 2 to 4 only	spill, any dose	task change
0.1	0.22	0.64	0.90	0.58	0.94	-0.007	0.002
0.2	0.35	0.64	0.83	0.65	0.91	-0.006	-0.001
0.4	0.38	0.64	0.90	0.94	0.72	-0.007	0.004
0.8	0.41	0.65	0.74	0.74	0.76	0.002	0.002
1.6	0.42	0.65	0.87	0.82	0.90	0.015	-0.006
split-anti	0.38	0.65	0.88	0.84	0.92	0.025	0.000

The pooled pull by temperature of the pool over slices, seed 701, with `split-anti` at the same seed below. α at the example answers of `difference` probe contexts at block 1 and block 4. Removal at the full dose of the edit at every slice, then on block 1 alone and on blocks 2 to 4 alone. The spill and task change are as in E1. The first line is `pool-slices`, whose pool over slices is at the recipe τ .

Every run on this ladder is within the criterion at every dose, and the task change is small. The largest spill rises a little at the two softest temperatures, toward that of `split-anti` at this seed. Removal under the full edit is 0.74 to 0.90 with no trend across the temperatures, and the edit on block 1 alone or on blocks 2 to 4 alone moves around from one run to the next, also with no trend. With one run per temperature, differences of this size are within the spread between seeds that `pool-slices` and `deep` at a hold of 0.2 showed in E1, so they do not show an effect of the temperature.

Discussion

The high hold everywhere had two effects, at different slices. On the embedding and block 1 it keeps the stand-ins off e_1 , which fits where the spill came from; on the pulled blocks it presses on the alignment at the fitting example answers, which fits where the removal went. Splitting the weight keeps the first and lifts the second, as the second review expected.

Stand-in. Something token-level that satisfies the pull where no contextual concept can...

Spill. The largest net drop in score on another op under the edit....

Alignment. How far a state points along e_1 ...

Fitting example. An example whose answer is one that difference could give on its operands, whatever op the context is under.

Removal. How much of difference the edit takes out...

The one `split-anti` seed past the criterion spills on the same op where `deep` spilled least, and a little more of block 1 and the table sit on e_1 than at the high hold everywhere. So the split may trade a small amount of selectivity for the removal it gives back. If that trade is typical, a hold a little above 0.2 on the first two slices, or the hold of 0.2 with a higher one on block 1, would be the place to look.

Where the example answers aligned early, the pooled pull settled on block 4 and the result was much like the split. Where the query answer got ahead, the pull settled there alone, a position that is in every labeled context but comes after the prediction it would matter for. As a recipe the pooled pull would need the query answer left out of its pool, and even then it would make the anchor depend on which position aligns first.

On the one seed tried, a softer pool over slices made the pooled pull look like the split: blocks 1 to 4 all aligned, with block 4 still the furthest along. So the softer pool takes away the concentration over slices, which was the point of pooling over them, and on this seed what remained was close to the per-slice pull of `split-anti`.

Glossary

Alignment

α

How far a state points along e_1 : the cosine between the state and e_1 , which is 1 on the anchor direction and 0 off it.

Anti-subspace term

The companion to the pull: the mean of α^2 over every position and slice it acts on, which asks the states as a whole to stay off e_1 . Its weight starts at 0.25 and anneals to a level it then holds, the hold.

Fitting example

An example whose answer is one that `difference` could give on its operands, whatever op the context is under.

Stand-in

Something token-level that satisfies the pull where no contextual concept can: lightness on e_1 in the color embedding table (the lean), or a syntax token embedding on e_1 (a latch).

Removal

How much of `difference` the edit takes out: the net drop in its score under the edit, as a share of the way from the clean score to the target null.

Spill

The largest net drop in score on another op under the edit. The selectivity criterion is a spill of at most 0.02.

Target null

The score an ideal predictor would get on `difference` contexts if it had lost that op and answered from the remaining ones.